



Merit Scan

THESIS DEFENSE

MERIT SCAN

QR-Based Web System for Attendance and
Merit-Demerit Management

Alexandra Acedo • 4th Year - BSCS
La Consolacion College – Biñan

May 12, 2026

Teacher-Friendly Preview

Preview Only

First Screen

Today's attendance
28/30

Pending checks
2

Reports ready
Yes

Account Screens

Admin Dashboard,
scanner, approvals

Student
QR card, status,
notices



Modernizing Academic Management at LCC Biñan

Current Challenge

- Manual attendance and discipline tracking are slow and inefficient.
- Manual records are prone to human errors and data loss.
- Lack of integrated behavioral monitoring and centralized reporting.

The Solution: Merit Scan

- QR-based web platform for real-time attendance tracking.
- Automated updating of student merit and demerit records.
- Centralized system for alerts and behavioral monitoring.

**Transforms
attendance
tracking into a
comprehensive
management
system.**



Introducing Merit Scan: The Digital Solution

General Objective

To design and implement a secure QR-based attendance and merit tracking system that improves discipline monitoring and institutional communication.

Specific Objectives

Record attendance using unique QR codes for real-time data capture.

Provide real-time dashboards for monitoring attendance and behavioral patterns.

Automate merit/demerit updates and send alerts for early intervention.

Ensure system scalability and secure, role-based data storage.



Defining the System Boundaries

System Scope

- QR-based attendance and merit/demerit tracking.
- Role-based dashboards for Master, Admin, and Students.
- Automated alerts for absences and tardiness.
- Institutional announcements and event management.
- Exportable reports and centralized data logs.

Delimitations

- Requires a stable internet connection (No offline mode).
- Web-based platform only (No dedicated mobile app).
- Excludes biometric authentication (Fingerprint/Face ID).
- No integration with external LMS or financial systems.
- Limited to attendance and discipline management.



Beyond Simple Presence Tracking

The Research Gap

- Most current QR solutions focus exclusively on attendance validation.
- Existing systems lack integration with behavioral and discipline monitoring.
- Reactive approach: Data is recorded but not used for immediate intervention.

The Merit Scan Advantage

- Unified Monitoring: Integrates attendance capture with real-time merit/demerit tracking.
- Behavioral Analytics: Monitors student discipline and participation patterns.
- Proactive Intervention: Automated threshold detection for immediate response.

Moving from
"Presence" to
"Performance and
Participation".



Built for Stability and Scalability

Development Model

- Prototype Model: Iterative design and refinement based on stakeholder feedback.
- Stages:
 - Requirement Gathering
 - Quick Design & Prototyping
 - Customer Evaluation
 - Refining & Final Release

Technical Stack

HTML5

CSS3

ReactJS

PHP

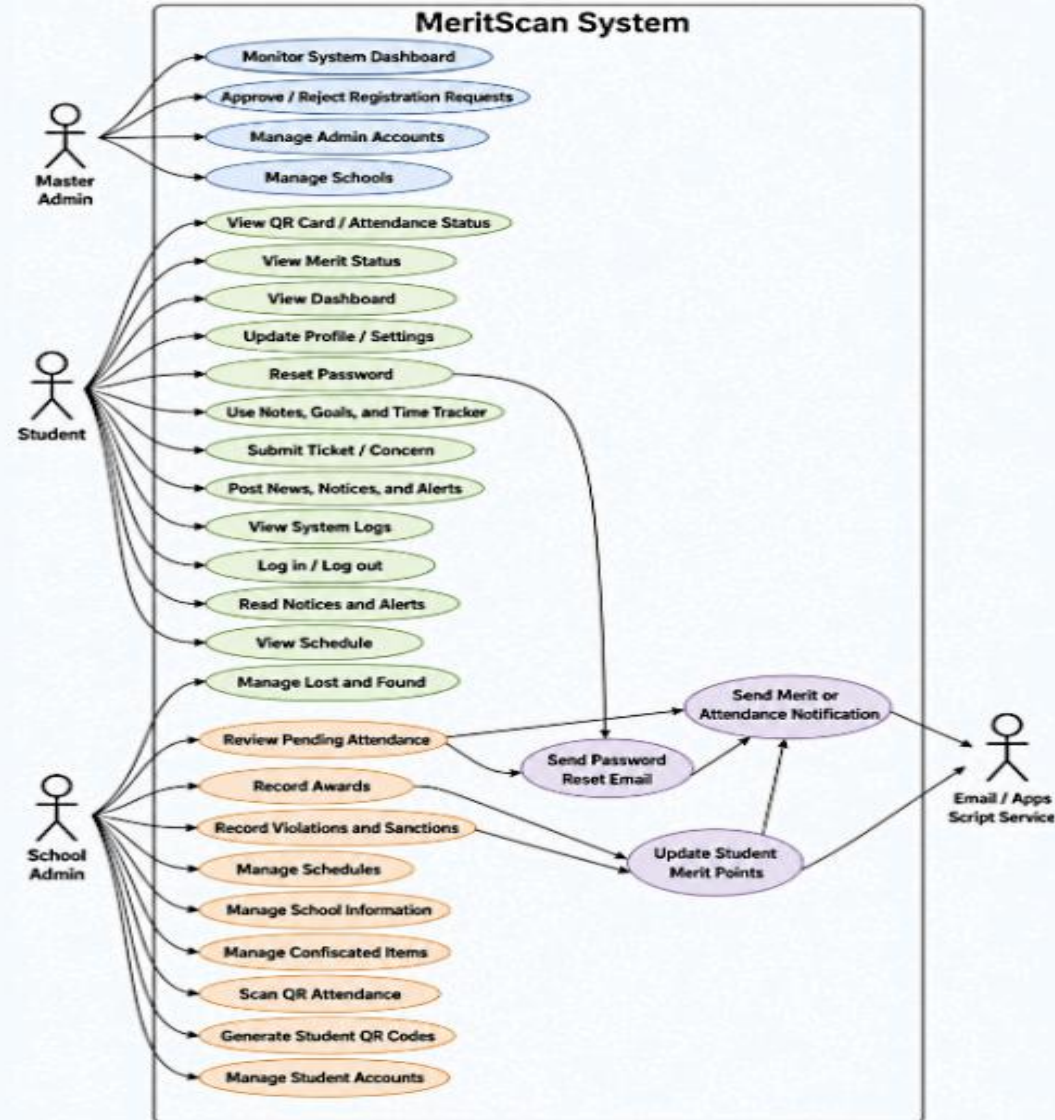
MySQL

Extra-Mile Feature

- OpenCV (Python): Smart uniform recognition for automated compliance checking.

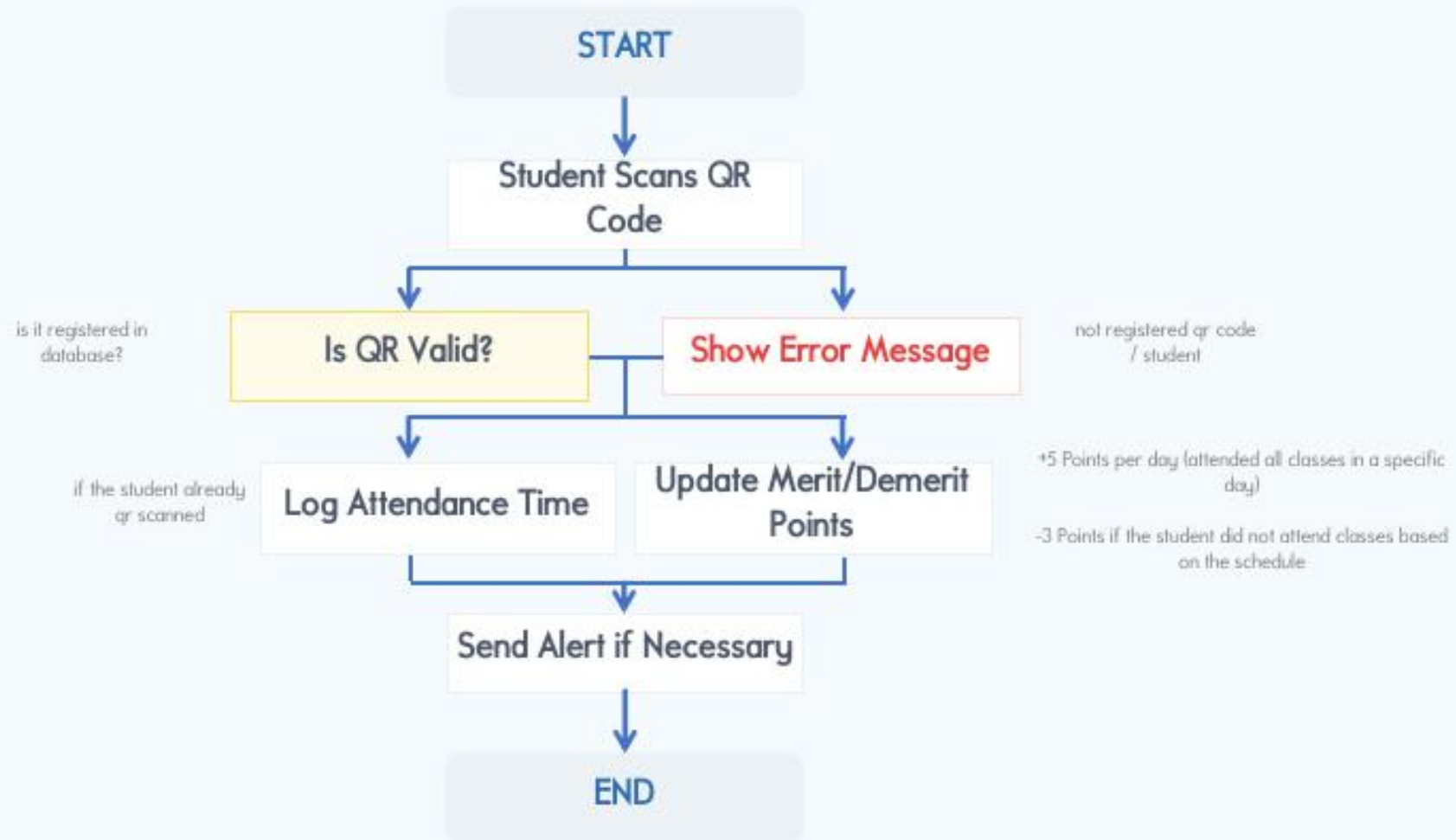


System Architecture: Use Case Diagram



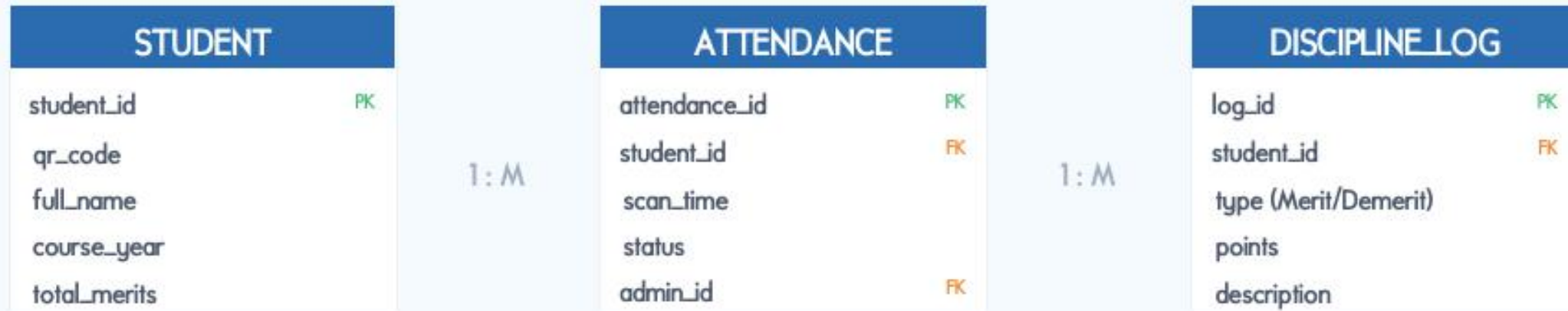


Process Logic: Attendance Recording





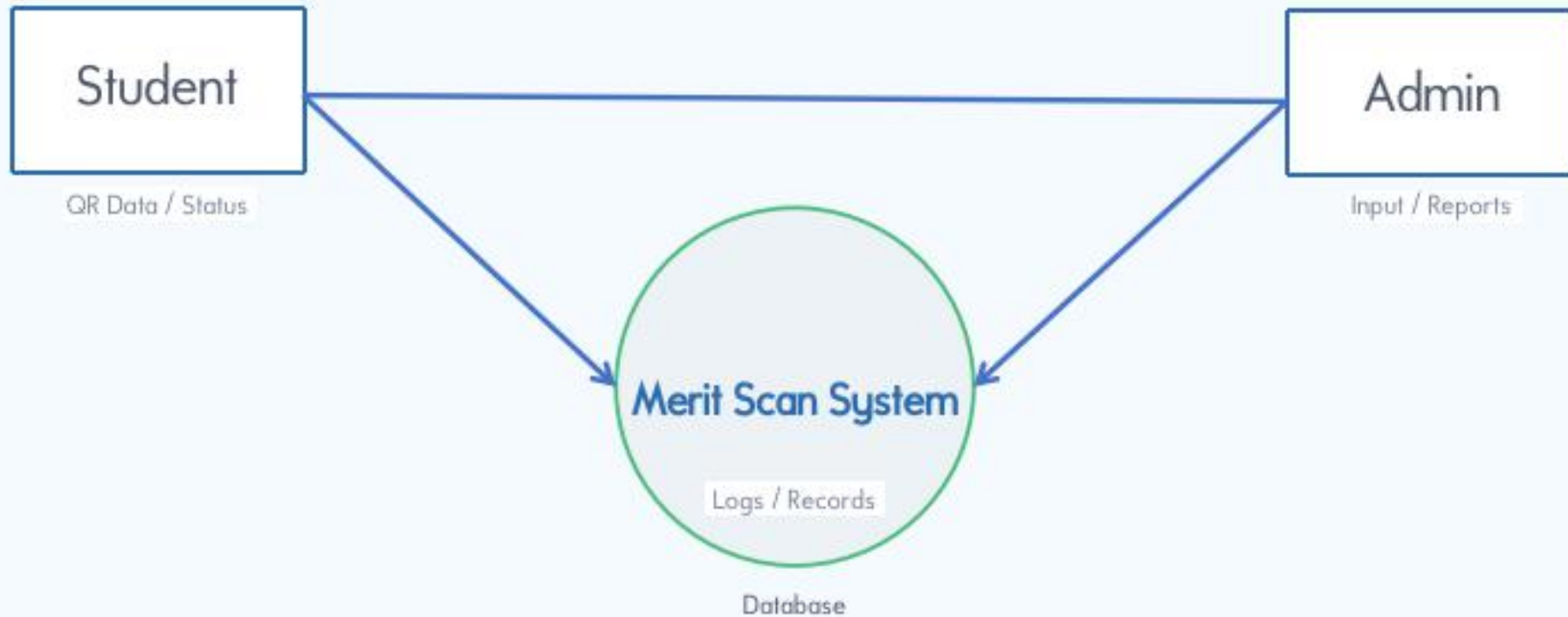
Database Design: Entity Relationship Diagram



Simple form of Database Design



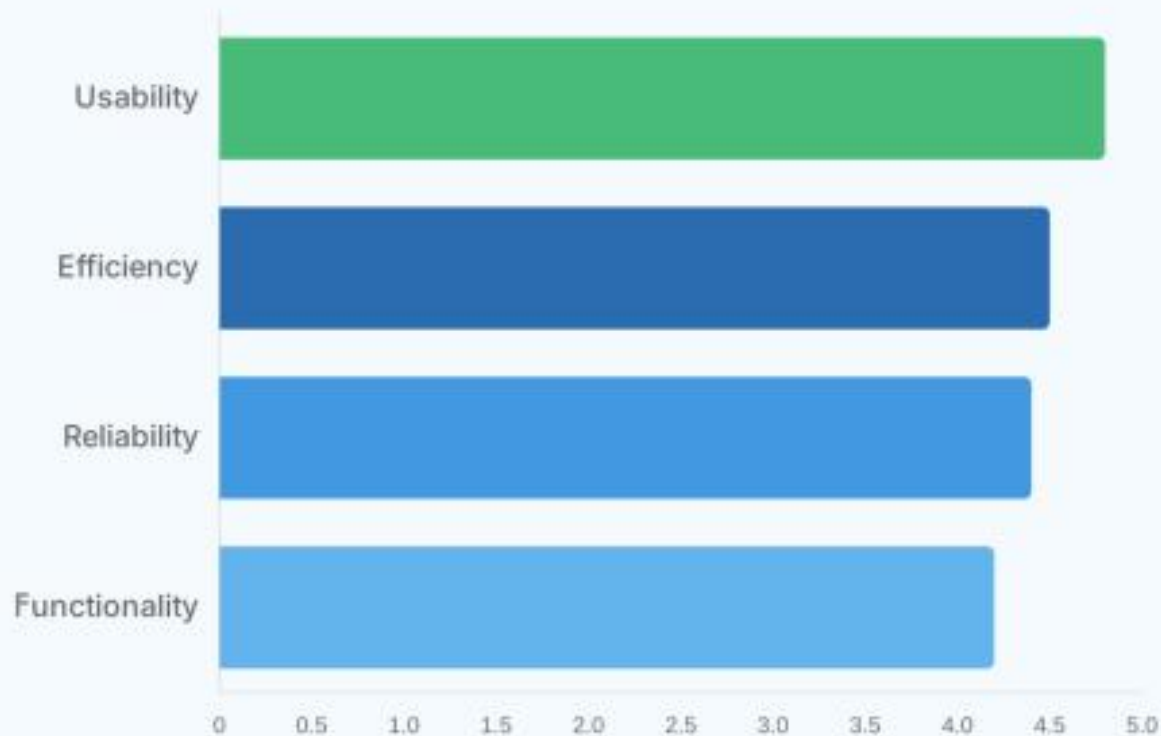
Data Flow Diagram: Level 0





Evaluation Results (ISO 9126)

Performance Metrics



Key Insights

Usability (4.8/5.0)

Strongest area; users found the interface intuitive and easy to learn for all roles.

Efficiency (4.5/5.0)

System responds quickly and performs functions without excessive resource needs.

Overall Interpretation

Highly suitable for implementation in a college-level academic environment.



Modernizing the Future of LCC Biñan

Project Summary

- Successfully transitioned manual roll calls to a centralized QR-based digital platform.
- Integrated attendance tracking with real-time merit and discipline management.

Conclusion

- The system significantly improves speed, accuracy, and transparency in student monitoring.
- Merit Scan addresses the core administrative challenges of La Consolacion College Biñan.

Recommendations

- Develop a dedicated mobile application for easier access and notifications.
- Implement SMS notification support as an alternative for users who may not always check their email.



Merit Scan

That's all, Thanks For Listening!

Thank you for your time
and attention.

QUESTIONS are welcome.